

4(a)	acceleration and displacement are in opposite directions	B1
4(b)(i)	$F = kx$ $= 8.0 \times (0.060 - 0.048)$ <i>or</i> $8.0 \times (0.060 + 0.048)$ <i>or</i> 8.0×0.012 <i>or</i> 8.0×0.108	M1
	$\Sigma F = (8.0 \times 0.012) - (8.0 \times 0.108) = 0.77\text{ N}$ <i>or</i> $\Sigma F = 0.864 - 0.096 = 0.77\text{ N}$	A1

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Question	Answer	Marks
4(b)(ii)	$a = \frac{F}{m}$ $= \frac{0.77}{0.25}$ $= 3.1 \text{ ms}^{-2}$	A1
4(b)(iii)	$a = -\omega^2 x$ $\omega = \sqrt{\frac{3.1}{0.048}}$ $\omega = 8.04$	C1
	$T = 2\pi / \omega$	C1
	$T = 2\pi / 8.04$ $= 0.78 \text{ s}$	A1
4(b)(iv)	(resultant) force halved and distance halved	B1
	same T	B1

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